Combustion and Flame xxx (2015) xxx–xxx

Contents lists available at ScienceDirect

Combustion and Flame

journal homepage: www.elsevier.com/locate/combustflame

Autoignition response of *n*-butanol and its blends with primary reference fuel constituents of gasoline

Kamal Kumar [a,*], Yu Zhang [a,1], Chih-Jen Sung [a], William J. Pitz [b]

[a] Department of Mechanical Engineering, University of Connecticut, Storrs, CT 06269, USA
[b] Lawrence Livermore National Laboratory, Livermore, CA 94551, USA

A R T I C L E I N F O

Article history:
Received 27 October 2014
Received in revised form 18 February 2015
Accepted 18 February 2015
Available online xxxx

Keywords:
n-Butanol
iso-Octane
n-Heptane
Autoignition
Fuel blend combustion

A B S T R A C T

We study the influence of blending *n*-butanol on the ignition delay times of *n*-heptane and *iso*-octane, the primary reference fuels for gasoline. The ignition delay times are measured using a rapid compression machine, with an emphasis on the low-to-intermediate temperature conditions. The experiments are conducted at equivalence ratios of 0.4 and 1.0, for a compressed pressure of 20 bar, with the temperatures at the end of compression ranging from 613 K to 979 K. The effect of *n*-butanol addition on the development of the two-stage ignition characteristics for the two primary reference fuels is also examined. The experimental results are compared to predictions obtained using a detailed chemical kinetic mechanism, which has been obtained by a systematic merger of previously reported base models for the combustion of the individual fuel constituents. A sensitivity analysis on the base, and the merged models, is also performed to understand the dependence of autoignition delay times on the model parameters.

1. Introduction

n-Butanol is an attractive fuel that can be used to displace gasoline in internal combustion engines. It is a bio-derived, renewable fuel whose use in transportation vehicles should lead to reduced emissions of carbon dioxide, a greenhouse gas [1]. It has a higher energy density than ethanol hence its use should also lead to greater vehicle mileage than ethanol [2]. Additionally, unlike ethanol, it could possibly be transported in existing fuel pipelines and infrastructure [3,4].

The possibility of using *n*-butanol as a blending agent or a replacement for gasoline has been the subject of several research efforts [5–9]. These efforts have explored both the physical as well as chemical compatibility of *n*-butanol as a substituent. There are several recent studies that have specifically looked at the physical properties of *n*-butanol from the perspective of a fuel additive. A study by Chotwichien et al. [5] reported a mixture consisting of 85% diesel, 10% biodiesel, and 5% butanol to be the most suitable as a diesohol (often referred to as a blend of alcohol and diesel fuel) in terms of phase stability and basic fuel properties according to the ASTM standards. In another study, nozzle cavitation and spray formation for *n*-butanol have been investigated by Aleiferis et al. [6]. Christensen et al. [7] have looked at several oxygenates, including butanol, for their utility as gasoline extenders. Their study identified cold start and drivability issues that could arise on account of the lower vapor pressure of some of the higher alcohols. Andersen et al. [8,9] have obtained vapor pressure and distillation curves for various gasoline-alcohol blends. In addition, they noted the need to understand the detailed physicochemical properties of gasoline-alcohol blends, prior to introducing them in the market. It is also important to make a mention of the internal combustion engine studies, which involve the in-cylinder blending of fuels having different autoignition properties. The differing autoignition trends of the two blend constituents are used to control the combustion phasing and excessive rates of heat release, and this method is referred to as the Reactivity Controlled Compression Ignition (RCCI). The in-cylinder blending in the RCCI concept leads to fuel reactivity gradients, which allows a control over the combustion duration. Further details on the RCCI concept and its implementation can be found in the recent studies of Reitz and co-workers [10–14].

Given the potential use of *n*-butanol as a gasoline substitute, there has been a surge in research related to *n*-butanol combustion. This has recently been reviewed by Sarathy et al. [15] who

* Corresponding author at: Department of Mechanical Engineering, University of Connecticut, 191 Auditorium Road, Storrs, CT 06269-3139, USA. Fax: +1 (860) 486 5088.
E-mail address: kamal@engr.uconn.edu (K. Kumar).
[1] Current address: Aramco Research Center – Detroit, 46535 Peary Ct., Novi, MI 48377, USA.

http://dx.doi.org/10.1016/j.combustflame.2015.02.014

reviewed fundamental combustion experiments and combustion chemistry for alcohols, including *n*-butanol. Fundamental experimental studies on *n*-butanol combustion have utilized well-controlled laboratory settings such as jet stirred reactors [16–21], shock tubes [22–31], laminar premixed and diffusion flames [19,32–47], flow reactor [48], ignition quality tester [49], and rapid compression machines [50–53]. Theoretical studies on obtaining the rate constants for hydrogen abstraction by radicals at different carbon atom sites [54–59], the decomposition kinetics of butanol derived radicals [60], and the thermochemistry of species relevant to *n*-butanol combustion [61,62] have also been carried out. Recently, a number of chemical kinetic models for *n*-butanol have been developed [22,63–65].

Moreover, there have been several applied studies investigating the effect of butanol addition to conventional transportation fuels in spark ignition [66–74], compression ignition [70,75–92], and homogeneous charge compression ignition (HCCI) engines [17,70,81,93–95]. By comparison, fundamental combustion studies on blends of *n*-butanol and a representative hydrocarbon component for transportation fuels are meager. Zhang et al. [96], preliminary version of the present paper, conducted autoignition experiments of binary blends of *n*-butanol and *n*-heptane in a rapid compression machine at an equivalence ratio of 0.4 and a compressed pressure of 20 bar with the compressed temperature varying from 700 K to 907 K. Speciation data on *n*-heptane and *n*-butanol blends at 700 K and at 9 atmospheres was reported by Karwat et al. [51]. In addition, ignition delay times for *n*-butanol and *n*-heptane blends have been measured by Zhang et al. [97], in which various fuel blend mixtures diluted with argon behind reflected shock waves were investigated in the temperature range of 1200–1500 K for 2 and 10 atmospheres. The laminar flame speeds of *n*-butanol/*iso*-octane mixtures were measured by Zhang et al. [98], showing that addition of *n*-butanol to *iso*-octane increased the laminar flame speed. Using an ignition quality tester, the derived cetane number test for blends of butanol isomers and ethanol conducted by Haas et al. [49] concluded that the ignition trends for primary and secondary alcohols were quite similar. Haas et al. [49] also observed that while *t*-butanol was the least reactive as a neat component, its blends with *n*-heptane or a diesel fuel were more reactive than the other alcohol blends for most mixtures. The observations in the study by Haas et al. [49] were attributed to relative C–H bond strengths of the methyl groups in *t*-butanol. More efforts toward understanding combustion characteristics for *n*-butanol blending with other hydrocarbon fuels are clearly warranted.

Since *n*-butanol is a likely replacement for gasoline, it is vital to acquire fundamental data on *n*-butanol and PRF constituent blends under engine relevant conditions. The current work is fundamental in nature and investigates the influence of blending *n*-butanol with two representative gasoline components: *n*-heptane and *iso*-octane. These two hydrocarbons are primary reference fuels for gasoline that define the knock resistance of gasoline through the octane index. They are also often used in various surrogate mixtures to represent gasoline fuels, where *n*-heptane is representative of the *n*-alkane chemical class and *iso*-octane is representative of the *iso*-alkane class in gasoline. In the present work, conditions relevant to internal combustion engines, including total fuel concentration, pressure, and temperature, are investigated. The temperature range investigated includes the negative temperature coefficient (NTC) region where two-stage ignition is often observed for hydrocarbon fuels. The present study is part of our effort to systematically study the oxidation of *n*-butanol-blended mixtures. Our aim is to start with binary fuel mixtures, and then move further into more complicated, multi-component fuel blends. The primary goal of the present work is to explore the autoignition behavior of the fuel mixtures of *n*-butanol/*n*-heptane and *n*-butanol/*iso*-octane as a function of the blending ratios between the two compounds under the physical conditions relevant to those in HCCI engines, i.e., ultra fuel-lean, low-to-intermediate temperatures, and elevated pressures. It is also our objective to build up a fundamental experimental database for the oxidation of *n*-butanol-blended fuel mixtures. In addition, experimental results are simulated using a detailed reaction mechanism constructed via a systematic merger of base models for the individual fuel constituents available in the literature.

2. Experimental specifications

The autoignition experiments have been conducted using a rapid compression machine (RCM), with provisions for pre-heating. This apparatus compresses a fixed mass of premixed fuel–air mixture from predetermined initial conditions to the desired high pressure/temperature conditions. A schematic and description of the RCM employed can be found in [99,100]. The pressure of the compressed charge is recorded both during and after the compression stroke, and is used to determine the pressure time history, which also incorporates the observed ignition response. The initial and the end-of-compression conditions of pressure and temperature can be related using the adiabatic compression relation given by:

$$\int_{T_0}^{T_C} \frac{\gamma}{\gamma-1}\frac{dT}{T} = \ln\left(\frac{P_C}{P_0}\right).$$

Here P_C and T_C are respectively the compressed charge pressure and temperature at the end of compression, P_0 and T_0 are respectively the measured initial pressure and temperature of the homogeneous charge, and γ is the specific heat ratio of the charge that varies with temperature. A Kistler 6125B thermal-shock resistant pressure transducer (uncoated) with a 5010B charge amplifier is used to measure the dynamic pressure within the reaction chamber. The maximum sensitivity shift for the pressure transducer as stated by the manufacturer is ±2%. Using the approximation $\Delta T/T = \frac{\gamma-1}{\gamma} \times \Delta P/P$ for compression of argon, we estimate $\Delta T/T = \pm 0.8\%$. This corresponds to a temperature uncertainty of approximately ±8 K for a compressed temperature of a 1000 K. The pressure signal is acquired at a rate of fifty samples per millisecond. Our earlier studies provide further details on operation of the RCM [99,100] as well as the experimental procedure for alcohols [52,53,101–103], neat hydrocarbon fuels [100,104,105], and fuel blends [106–109].

As noted earlier, the pressure time record is used to infer the ignition delay time. For the RCM experiments, the ignition delay time is determined with reference to the end of compression. The maximum rate of pressure rise in the post-compression pressure record is defined as the onset of ignition, both for the first stage ignition as well as the total (hot) ignition (cf. [99,100]).

In this study, *n*-butanol is obtained from Sigma–Aldrich (99.9% purity) whereas *n*-heptane and *iso*-octane are obtained from Fisher Scientific (both 99.5% purity). A binary blend of the alcohol with any one of the hydrocarbons is used as the test fuel. The molar percentage of *n*-butanol in the binary fuel blend is denoted by the abbreviation R_b for this work. Most of the RCM experiments are conducted at an equivalence ratio of $\phi = 0.4$ and a compressed pressure of $P_C = 20$ bar. Additional dataset for stoichiometric blends at $P_C = 20$ bar is also carried out. Table 1 lists the compositions of the gas mixtures tested in this study, while the molecular structures of *n*-butanol, *n*-heptane, and *iso*-octane are given in Fig. 1 to facilitate chemical kinetic discussions in due course. The experimental results and the associated uncertainties for the ignition delay times for fuel-lean and stoichiometric mixtures are provided in Tables 2 and 3, respectively.

Table 1
Compositions tested in this study.

ϕ	R_b	*n*-Butanol	*n*-Heptane	Oxygen	Nitrogen	
		Molar percent				
0.4	0	0	0.758	20.8	78.4	
0.4	20	0.167	0.667	20.8	78.3	
0.4	40	0.370	0.555	20.8	78.3	
0.4	60	0.624	0.416	20.8	78.2	
0.4	80	0.949	0.237	20.8	78.1	
ϕ	R_b	*n*-Butanol	*iso*-Octane	Oxygen	Nitrogen	
		Molar percent				
0.4	0	0	0.668	20.9	78.5	
0.4	60	0.581	0.387	20.8	78.2	
0.4	100	1.381	0	20.7	77.9	
ϕ	R_b	*n*-Butanol	*n*-Heptane	Oxygen	Nitrogen	Argon
		Molar percent				
1.0	0	0	0.545	6.0	22.6	70.9
1.0	20	0.120	0.480	6.0	22.6	70.8
1.0	40	0.267	0.400	6.0	22.6	70.8
1.0	60	0.450	0.300	6.0	22.6	70.7
1.0	80	0.686	0.171	6.0	22.6	70.6
1.0	100	1.000	0	6.0	22.6	70.4
ϕ	R_b	*n*-Butanol	*iso*-Octane	Oxygen	Nitrogen	Argon
		Molar percent				
1.0	0	0	1.200	15.0	56.4	27.4
1.0	20	0.268	1.071	15.0	56.4	27.3
1.0	40	0.606	0.909	15.0	56.4	27.1
1.0	60	1.047	0.698	15.0	56.4	26.9
1.0	80	1.644	0.411	15.0	56.4	26.5
1.0	100	2.500	0	15.0	56.4	26.1

3. Computational specifications

The experimental results in this study are simulated using a detailed chemical kinetic scheme for *n*-butanol and the primary reference fuels for gasoline, namely, *n*-heptane and *iso*-octane. This mechanism, derived from a systematic merger of previously reported schemes for *n*-butanol by Sarathy et al. [65], *n*-heptane by Karwat et al. [51], and LLNL *iso*-octane (Version 3) [110–113], consists of 2385 species and 9515 reactions.

The rapid compression machine experiments are simulated based on zero-dimensional homogeneous kinetics calculations using CHEMKIN® PRO [114] and CHEMKIN-III [115] with SENKIN [116]. The initial conditions corresponding to P_0, T_0, and reactant species mole fractions are specified as inputs in the simulation.

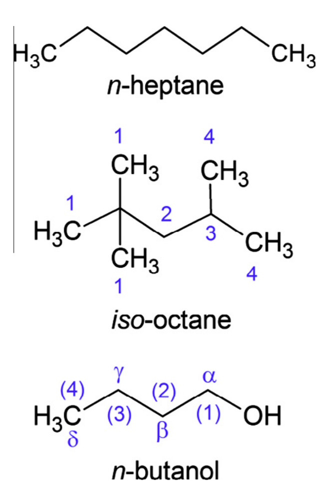


Fig. 1. Molecular structure of fuel components tested in this study.

Furthermore, the reactor volume in the simulation is varied as a function of time to mimic the entire experimental compression stroke. Additionally, heat losses during compression stroke as well as the post compression phase are empirically accounted for in the simulations, as detailed in e.g. [52,53,99,101–105].

4. Results and discussion

Figure 2 shows selected experimental pressure traces from the oxidation of the three neat components, and a corresponding Arrhenius plot for the total ignition delays over the temperature range investigated. Moreover, the corresponding simulated results are also included in Fig. 2 for comparison. Figure 2(d) gives all the total ignition delays for the three fuel components considered, for an equivalence ratio (ϕ) of 0.4 and a compressed charge pressure (P_C) of 20 bar. These are new ignition delay measurements made for the three neat components at the specific conditions, and hence are additional targets for model validation. As can be seen, total ignition delay decreases monotonically with an increase in compressed temperature (T_C) for each component, and that the *n*-heptane and *iso*-octane fuels respectively describe the lower and upper limits, with the *n*-butanol in-between, but closer to *iso*-octane. The ignition delay trends for the three neat fuels are found to be in agreement with the known octane number, (*Research Octane Number* +*Motor Octane Number*)/2, in that the values for these fuels [71,117] are *n*-heptane (0), *n*-butanol (86), and *iso*-octane (100). In Fig. 2(a)–(c), we note the presence of a distinct two-stage ignition behavior for *n*-heptane and a lack of two-stage behavior (i.e. single-stage ignition) for *iso*-octane and *n*-butanol, within the temperature and pressure range covered herein.

In general, the combined chemical kinetic model predicts the autoignition responses for all three neat components reasonably well, especially for *n*-heptane. As shown in Fig. 2(b) and (d) for *n*-heptane, both the experimental total ignition delays and pressure traces, including the first-stage ignition delays and the first-stage pressure rises, are well captured by the simulations. The differences in the shapes of the pressure traces between the simulations and experiments could be attributed to multidimensio nal and crevice flow effects not captured by a single-zone model. It is also noted that for the temperature range investigated the chemical kinetic model predicts faster and slower autoignition reactivity for *n*-butanol and *iso*-octane, respectively, while the discrepancy between the simulated and experimental results narrows down at higher compressed temperatures. Further efforts, both experimental and computational, are required to identify this discrepancy. The general features of the experimental pressure traces for *n*-butanol and *iso*-octane are also seen in the simulated profiles, as shown in Fig. 2(a) and (c), respectively. As such, the comparison demonstrated in Fig. 2 indicates the robustness of the low-temperature sub-mechanism for each neat component included in the detailed kinetic scheme.

It is well-established that two-stage ignition behavior is initiated through the dissociation of ketohydroperoxides, which are produced via a sequence of low temperature oxidation steps [118]:

$$R + O_2 + M = RO_2 + M \tag{1}$$

$$RO_2 = QOOH \tag{2}$$

$$QOOH + O_2 = O_2QOOH \tag{3}$$

$$O_2QOOH \rightarrow \text{ketohydroperoxide} + OH. \tag{4}$$

Among these low temperature reactions, the key steps are those isomerization reactions that involve internal H-atom transfers by

Please cite this article in press as: K. Kumar et al., Combust. Flame (2015), http://dx.doi.org/10.1016/j.combustflame.2015.02.014

Table 2
Experimental results for fuel-lean blends (P_C = 20 bar, ϕ = 0.4).

n-Butanol (Molar percent)	*n*-Heptane (Molar percent)	*iso*-Octane (Molar percent)	O_2 (Molar percent)	N_2 (Molar percent)	T_C (K)	R_b	τ (ms)	$\pm\tau$ (ms)
Ignition delays for blends of n-butanol with n-heptane and iso-octane								
		0.668	20.9	78.5	979	0	3.27	0.29
		0.668	20.9	78.5	965	0	5.28	0.26
		0.668	20.9	78.5	953	0	7.38	0.63
		0.668	20.9	78.5	925	0	12.22	0.44
		0.668	20.9	78.5	915	0	17.44	0.40
		0.668	20.9	78.5	904	0	22.56	0.69
		0.668	20.9	78.5	887	0	27.60	0.88
		0.668	20.9	78.5	873	0	37.75	1.46
		0.668	20.9	78.5	860	0	44.46	1.19
0.581		0.387	20.8	78.2	948	60	5.46	0.27
0.581		0.387	20.8	78.2	933	60	7.62	0.20
0.581		0.387	20.8	78.2	918	60	10.26	0.25
0.581		0.387	20.8	78.2	900	60	14.82	0.59
0.581		0.387	20.8	78.2	891	60	19.50	0.36
0.581		0.387	20.8	78.2	880	60	23.88	0.78
0.581		0.387	20.8	78.2	866	60	30.32	0.70
0.581		0.387	20.8	78.2	857	60	41.90	0.70
0.581		0.387	20.8	78.2	841	60	50.96	1.26
0.581		0.387	20.8	78.2	830	60	59.55	1.17
0.581		0.387	20.8	78.2	814	60	65.82	1.17
0.581		0.387	20.8	78.2	802	60	71.04	1.54
0.581		0.387	20.8	78.2	789	60	76.89	0.88
1.381			20.7	77.9	962	100	2.30	0.15
1.381			20.7	77.9	948	100	3.01	0.30
1.381			20.7	77.9	933	100	3.78	0.21
1.381			20.7	77.9	917	100	6.18	0.19
1.381			20.7	77.9	904	100	9.27	0.35
1.381			20.7	77.9	891	100	12.66	0.55
1.381			20.7	77.9	877	100	16.62	0.47
1.381			20.7	77.9	866	100	20.70	0.50
1.381			20.7	77.9	854	100	26.06	0.40
1.381			20.7	77.9	837	100	32.76	1.16
1.381			20.7	77.9	828	100	40.14	1.57
1.381			20.7	77.9	809	100	46.10	0.77
1.381			20.7	77.9	800	100	53.84	1.15
0	0.758		20.8	78.4	735	0	1.96	0.07
0	0.758		20.8	78.4	719	0	2.77	0.21
0	0.758		20.8	78.4	701	0	4.10	0.23
0	0.758		20.8	78.4	682	0	6.54	0.22
0	0.758		20.8	78.4	665	0	10.97	0.24
0	0.758		20.8	78.4	655	0	16.00	0.66
0	0.758		20.8	78.4	646	0	23.58	0.33
0	0.758		20.8	78.4	637	0	30.91	0.98
0	0.758		20.8	78.4	619	0	85.09	2.72
0	0.758		20.8	78.4	613	0	120.42	2.09
0.167	0.667		20.8	78.3	778	20	2.18	0.05
0.167	0.667		20.8	78.3	759	20	2.34	0.04
0.167	0.667		20.8	78.3	741	20	2.84	0.05
0.167	0.667		20.8	78.3	726	20	3.79	0.21
0.167	0.667		20.8	78.3	713	20	5.30	0.22
0.167	0.667		20.8	78.3	700	20	7.31	0.20
0.167	0.667		20.8	78.3	691	20	10.11	0.60
0.167	0.667		20.8	78.3	676	20	16.18	0.71
0.167	0.667		20.8	78.3	665	20	20.46	0.76
0.167	0.667		20.8	78.3	655	20	46.96	0.29
0.167	0.667		20.8	78.3	645	20	73.48	0.57
0.167	0.667		20.8	78.3	638	20	94.42	2.01
0.370	0.555		20.8	78.3	803	40	2.81	0.03
0.370	0.555		20.8	78.3	790	40	2.70	0.10
0.370	0.555		20.8	78.3	779	40	2.88	
0.370	0.555		20.8	78.3	759	40	3.41	0.19
0.370	0.555		20.8	78.3	744	40	3.94	0.05
0.370	0.555		20.8	78.3	728	40	5.48	0.28
0.370	0.555		20.8	78.3	713	40	7.92	0.08
0.370	0.555		20.8	78.3	703	40	11.02	0.20
0.370	0.555		20.8	78.3	702	40	11.63	1.26
0.370	0.555		20.8	78.3	691	40	16.83	0.64
0.370	0.555		20.8	78.3	677	40	29.76	0.37
0.370	0.555		20.8	78.3	665	40	48.26	1.05
0.370	0.555		20.8	78.3	655	40	93.98	2.26
0.370	0.555		20.8	78.3	647	40	174.25	2.72
0.624	0.416		20.8	78.2	836	60	4.66	0.57
0.624	0.416		20.8	78.2	800	60	4.92	0.13

Table 2 (*continued*)

n-Butanol (Molar percent)	*n*-Heptane (Molar percent)	*iso*-Octane (Molar percent)	O_2 (Molar percent)	N_2 (Molar percent)	T_C (K)	R_b	τ (ms)	$\pm\tau$ (ms)
0.624	0.416		20.8	78.2	777	60	4.23	0.27
0.624	0.416		20.8	78.2	759	60	4.63	0.11
0.624	0.416		20.8	78.2	740	60	5.48	0.24
0.624	0.416		20.8	78.2	726	60	8.84	0.31
0.624	0.416		20.8	78.2	714	60	13.70	1.10
0.624	0.416		20.8	78.2	712	60	11.81	0.11
0.624	0.416		20.8	78.2	703	60	20.22	0.77
0.624	0.416		20.8	78.2	690	60	33.52	1.13
0.624	0.416		20.8	78.2	676	60	54.72	1.59
0.624	0.416		20.8	78.2	667	60	111.66	4.44
0.949	0.237		20.8	78.1	908	80	5.34	0.07
0.949	0.237		20.8	78.1	870	80	9.14	0.17
0.949	0.237		20.8	78.1	856	80	10.33	0.56
0.949	0.237		20.8	78.1	830	80	10.16	1.23
0.949	0.237		20.8	78.1	797	80	10.34	0.24
0.949	0.237		20.8	78.1	779	80	9.49	
0.949	0.237		20.8	78.1	760	80	10.49	0.15
0.949	0.237		20.8	78.1	742	80	13.05	0.21
0.949	0.237		20.8	78.1	729	80	20.04	1.01
0.949	0.237		20.8	78.1	726	80	18.57	0.27
0.949	0.237		20.8	78.1	717	80	28.73	1.09
0.949	0.237		20.8	78.1	702	80	46.19	0.92
0.949	0.237		20.8	78.1	693	80	80.65	2.97
0.949	0.237		20.8	78.1	685	80	101.56	5.12
0.949	0.237		20.8	78.1	679	80	169.24	13.84

Table 3
Experimental results for stoichiometric blends (P_C = 20 bar).

n-Butanol (Molar percent)	*n*-Heptane (Molar percent)	*iso*-Octane (Molar percent)	O_2 (Molar percent)	N_2 (Molar percent)	Ar (Molar percent)	T_C (K)	R_b	τ_1 (ms)	$\pm\tau_1$ (ms)	τ (ms)	$\pm\tau$ (ms)
Ignition delays for blends of n-butanol with n-heptane and iso-octane											
0	0.545		6.0	22.6	70.89	708	0	5.36	0.56	13.80	0.41
0.120	0.480		6.0	22.6	70.84	707	20	7.86	0.31	18.26	0.23
0.267	0.400		6.0	22.6	70.77	706	40	15.37	0.33	28.28	0.10
0.450	0.300		6.0	22.6	70.68	706	60	26.80	0.86	45.76	0.63
0.686	0.171		6.0	22.6	70.58	707	80	59.24	4.72	107.72	2.40
0.000		1.20	15.0	56.4	27.40	752	0	6.36	1.13	46.75	1.65
0.268		1.07	15.0	56.4	27.26	753	20	11.42	1.28	46.82	0.98
0.606		0.91	15.0	56.4	27.08	752	40	20.26	2.69	45.03	0.78
1.047		0.70	15.0	56.4	26.86	751	60	34.64	2.89	53.07	0.80
1.644		0.41	15.0	56.4	26.55	751	80	50.26	2.42	55.11	0.99

forming transition-state rings. Among the internal H-atom shift reactions, (1,5) H-atom shift (involving a six-membered ring transition state) is usually considered to have the most rapid reaction rate due to its low strain energy barrier and rather large pre-exponential *A*-factor.

The occurrence of both a two-stage ignition and NTC trend for ignition delays for the linear C_4 alkane has been previously reported [119,120]. Since *n*-butanol has a linear alkyl chain of four carbon atoms, one might expect it to exhibit a two-stage ignition behavior based on the above-mentioned low temperature oxidation sequence. However, no noticeable two-stage ignition behavior for this simplest C_4 alcohol is seen in this study. The suppression of the low temperature branching is mainly due to the effect of the hydroxyl group in alcohols. The reaction of O_2 with the radical site on the carbon next to the hydroxyl group (α-position) primarily leads to aldehyde + HO_2 [65] rather than chain branching as would occur without the presence of the hydroxyl group. As a result, low temperature chain branching and thus two-stage ignition behavior is suppressed in *n*-butanol oxidation.

Since C–H bonds at the α-position of alcohols (cf. Fig. 1) have the lowest bond dissociation energies among the C–H bonds in alcohols due to the presence of hydroxyl group [29], it is known that the reaction of α-hydroxyalkylperoxy radicals play a significant role in the low temperature oxidation of alcohols. The work of da Silva et al. [121] studied the α-hydroxylethyl + O_2 reactions via quantum chemical calculations. It was reported by the authors that the energy barriers for the dissociation of the α-hydroxylethylperoxy radical into acetaldehyde and hydroperoxy radical or vinyl alcohol and hydroperoxy radical through concerted elimination are considerably lower than those of isomerization reactions through intramolecular H-atom shift. da Silva et al. [121] noted that the short lifetime of α-hydroxylethylperoxy radical prevents the oxidation of the alkylhydroperoxide (QOOH) radical, responsible for NTC behavior in alkane autoignition. In their results, this lifetime starts to increase only moderately above 10 atm, so that very high pressures may be needed for the possibility of observing NTC behavior. Therefore, for the oxidation of *n*-butanol at low temperatures, it is reasonable to consider that the fast concerted HO_2 elimination reaction of α-hydroxylbutyl radical with O_2 would also dominate over the isomerization reactions through (1,5)H-atom transfers. This dominance of the fast concerted HO_2 elimination reaction leads to the lack of two-stage ignition and NTC behavior of ignition delay for *n*-butanol observed in this study.

Figures 3 and 4 summarize the total ignition delay trends for the blends of *n*-butanol with *n*-heptane and *iso*-octane, respectively. In Fig. 3(a), the behavior of the two pure components, *n*-heptane and *n*-butanol, bound the behavior of the binary blends

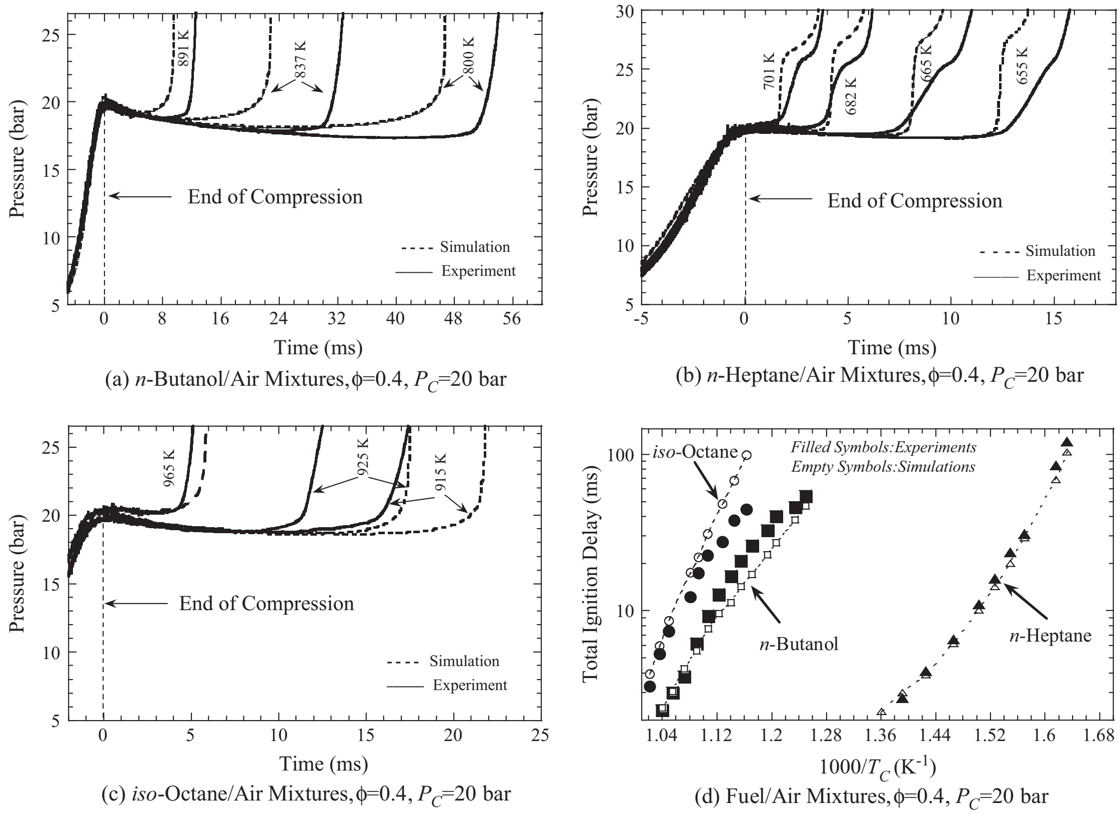


Fig. 2. Comparative experimental and simulated results for (a) pressure traces of *n*-butanol/air mixtures, (b) pressure traces of *n*-heptane/air mixtures, (c) pressure traces of *iso*-octane/air mixtures, and (d) Arrhenius plot of total ignition delays for pure components.

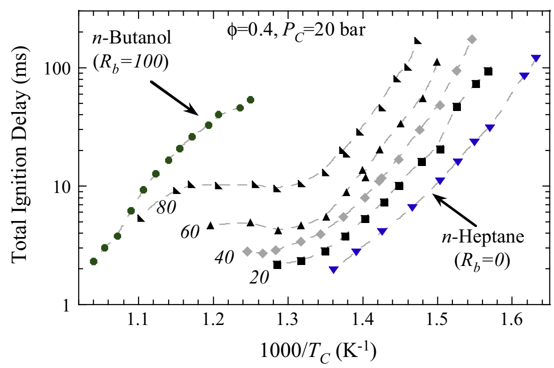
which fill the space in-between. Interestingly, *n*-butanol happens to fall in the fuel compatibility zone for gasoline while *n*-heptane is closer to diesel, as assessed by their respective octane and cetane numbers. The cetane number of *n*-heptane has been reported to be 54 [122], while the minimum specification of diesel cetane number can vary in the range of 40–51 [123]. The reported octane number for *n*-butanol using the $(\textit{Research Octane Number} + \textit{Motor Octane Number})/2$ method is 86 [71], which is comparable to gasoline. The effect of adding *n*-butanol to *n*-heptane is an increase in the total ignition delay. It is of interest to note that the blends also inherit the two-stage ignition behavior as observed for *n*-heptane, but not found in *n*-butanol as shown in Fig. 2(a). Figure 3(a) further demonstrates that no two-stage ignition or negative temperature coefficient (NTC) behavior has been observed from the oxidation of neat *n*-butanol for the conditions investigated. This result is consistent with the very high octane sensitivity of *n*-butanol $(RON - MON)$ of 14 [124]. As shown in Mehl et al. [125] for gasoline fuels, high sensitivity fuels do not exhibit NTC behavior. A 80–20 blend of *n*-butanol and *n*-heptane, i.e. $R_b = 80$, is observed to exhibit a pronounced two-stage ignition response (cf. Fig. 5), indicating that the presence of *n*-heptane in the binary mixture promotes chain branching at low temperatures appreciably. As previously mentioned, the first-stage ignition is initiated through the decomposition of ketohydroperoxides at low temperatures. In Fig. 3(a), it is also found that the change of total ignition delay with compressed temperature does not follow a monotonic trend for the R_b = 80 mixture, instead, NTC behavior is observed.

Another key observation from Fig. 3(a) is that the difference in total ignition delay between neat *n*-butanol and a mixture of *n*-butanol and *n*-heptane is temperature-dependent. The difference is more pronounced at low temperatures, while it becomes smaller as we traverse into the intermediate temperature range. Indeed, the total ignition delay curves appear to be merging for $R_b = 100$ and $R_b = 80$ at temperatures beyond 900 K. This temperature-dependent nature of the difference in total ignition delay between neat *n*-butanol and a mixture of *n*-butanol and *n*-heptane can be attributed mainly to the diminishing of two-stage ignition response for the *n*-butanol-*n*-heptane blend when the compressed temperature shifts from the low temperature range and enters the intermediate temperature range.

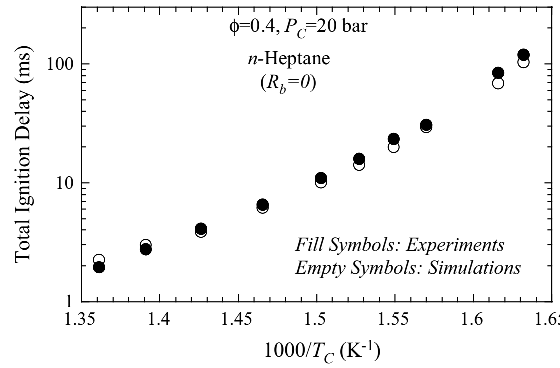
Figure 3(b)–(d) shows the comparative results for the experimental (filled symbols) and simulated (empty symbols) ignition delay times. It can be seen from Fig. 3(b) that there is a very good agreement between the simulated and experimental results for pure *n*-heptane. The mechanism is also able to fairly well reproduce the results for the fuel blend corresponding to 40% butanol, with a factor of 2–4 under-prediction for temperatures below 690 K, as shown in Fig. 3(c). There is good agreement for the case corresponding to a 80–20 blend of *n*-butanol and *n*-heptane. In particular, the simulation is able to accurately capture the plateau in the negative temperature coefficient range of 780–910 K, as shown in Fig. 3(d).

The ignition delay plot for the blends of *iso*-octane and *n*-butanol shown in Fig. 4 indicates that addition of *n*-butanol to *iso*-octane increases the reactivity, thereby shortening the total ignition delay times. This effect is in the opposite direction seen for addition of *n*-butanol to *n*-heptane. The relative magnitude of the influence is not as pronounced as that observed for *n*-heptane. It can be visually inferred from Fig. 4 that the total ignition delays for *n*-butanol are lower approximately by a factor of 2–3 as compared to pure *iso*-octane, over the temperature range investigated in this work.

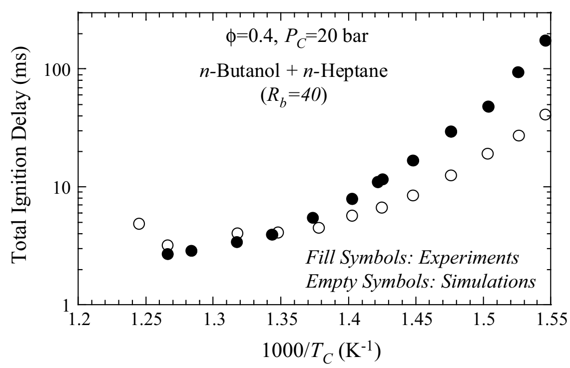
A comparison between the simulated and experimental results is shown in Fig. 4(b)–(d). It can be seen that the merged



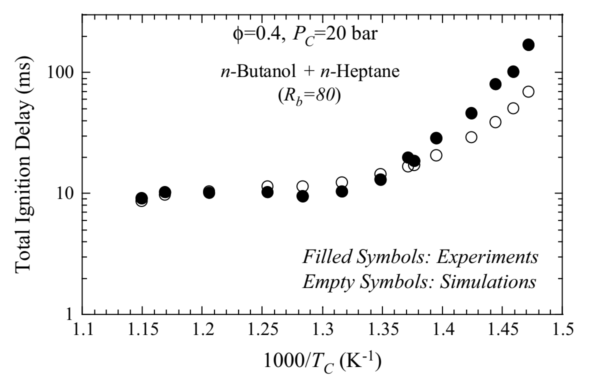
(a) Total Ignition Delays for *n*-Butanol/*n*-Heptane Blends in Air



(b) Total Ignition Delays for *n*-Heptane in Air

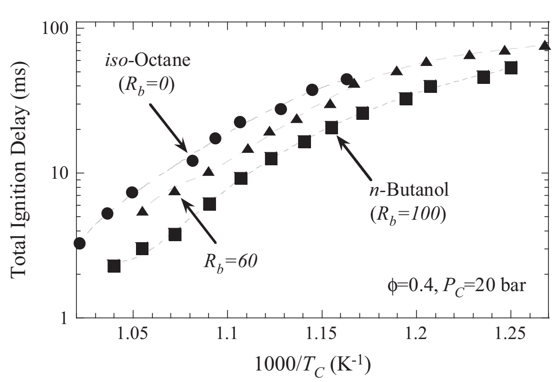


(c) Total Ignition Delays for *n*-Butanol/*n*-Heptane Blends in Air

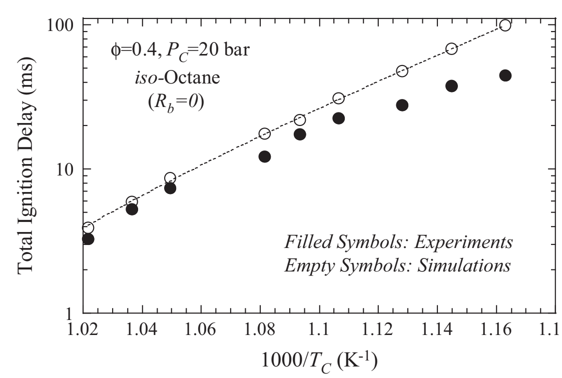


(d) Total Ignition Delays for *n*-Butanol/*n*-Heptane Blends in Air

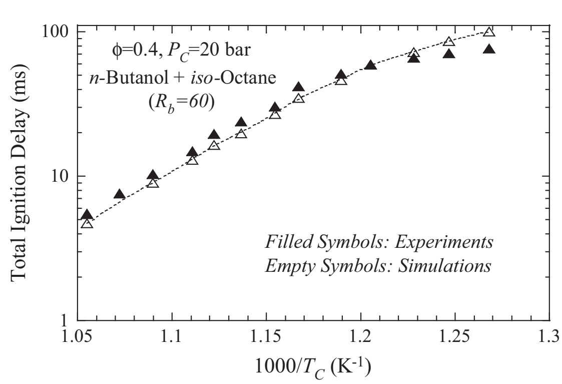
Fig. 3. Arrhenius plots of total ignition delays for *n*-butanol and *n*-heptane blends (cf. Table 1 for reactant molar percentages).



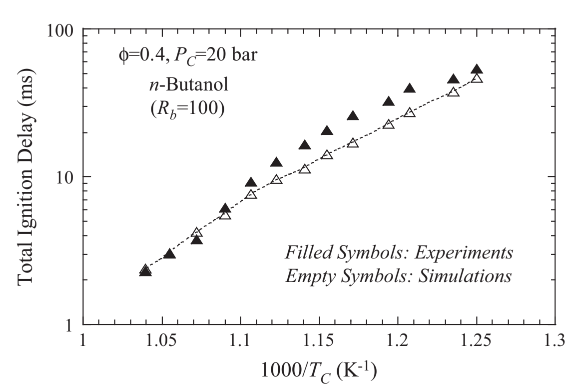
(a) Total Ignition Delays for *n*-Butanol/*iso*-Octane Blends in Air



(b) Total Ignition Delays for *iso*-Octane in Air

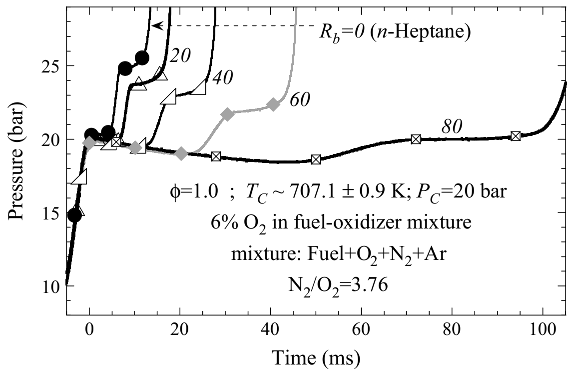


(c) Total Ignition Delays for *n*-Butanol/*iso*-Octane Blends in Air

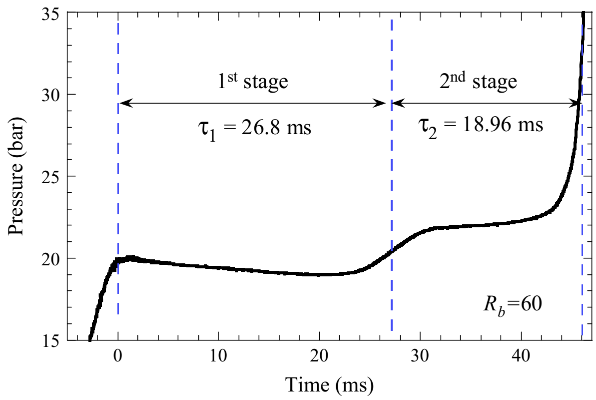


(d) Total Ignition Delays for *n*-Butanol in Air

Fig. 4. Arrhenius plots of total ignition delays for *n*-butanol and *iso*-octane blends (cf. Table 1 for reactant molar percentages).



(a) Reactive Pressure Traces for *n*-Butanol/*n*-Heptane Blends



(b) Illustration of First and Second Stage Ignition Delay

Fig. 5. Influence of blending ratio (R_b) on the two-stage ignition development for *n*-butanol and *n*-heptane blends at a fixed (P_C, T_C, ϕ).

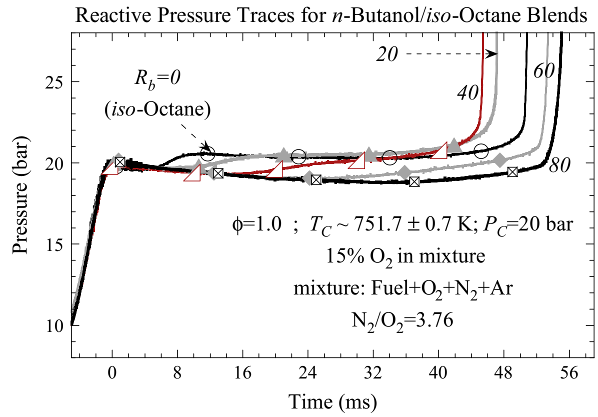


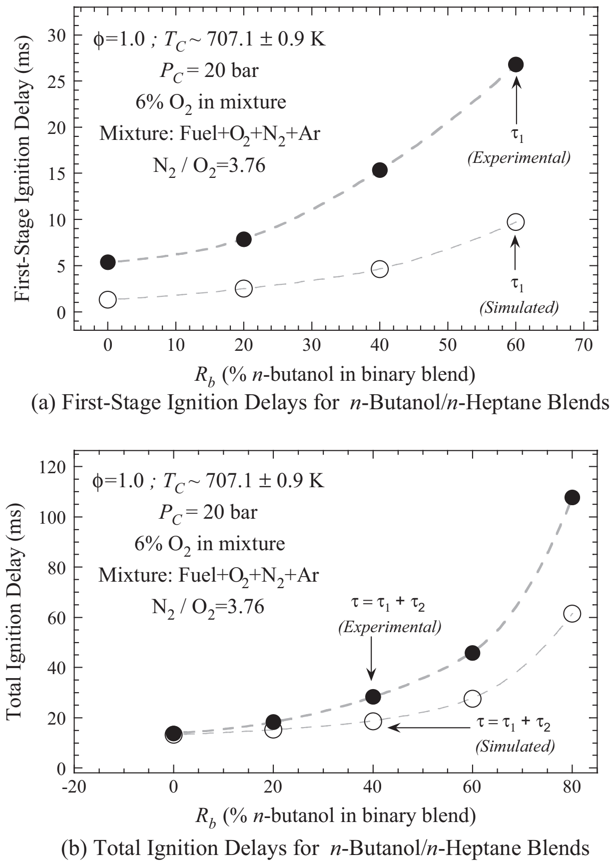
Fig. 6. Influence of blending ratio (R_b) on the two-stage ignition development for *n*-butanol and *iso*-octane blends at a fixed (P_C, T_C, ϕ).

mechanism slightly over-predicts (is slower than) the current experimental results for pure *iso*-octane, while slightly under-predicting (is faster than) the experimental results for pure *n*-butanol. As a result, there is a very good agreement between the simulated and experimental results for *n*-butanol and *iso*-octane blends, as shown in Fig. 4(c). It is important to note the merged mechanism accurately predicts the results in the high temperature range above 900 K for both neat components, and the deviations in predictions under low temperature conditions is within acceptable limits.

To further study the qualitative and quantitative characteristics of the two-stage ignition development process for alkane-alcohol blends, a separate set of experiments is conducted where, the effect of varying R_b on the ignition response is studied at fixed P_C and T_C. Stoichiometric binary fuel blends, with a variable proportion of *n*-butanol (R_b), are prepared with argon plus nitrogen as the diluent. The proportion of diluent for each fuel blend is chosen with the twin objectives of amplifying the first-stage pressure-rise and observing the ignition response within a similar timescale of 2–100 ms. The experimental pressure traces for the stoichiometric *n*-butanol/*n*-heptane and *n*-butanol/*iso*-octane blends are shown in Figs. 5(a) and 6, respectively. Figure 5(b) is used to illustrate the durations corresponding to the first and second stage ignition delays for the case corresponding to $R_b = 60$ in Fig. 5(a). Note that though the nominal compressed pressures are the same for Figs. 5 and 6 ($P_C = 20$ bar), the resulting compressed temperature and the molar percentage of oxygen in the mixture are different. The *n*-butanol/*n*-heptane blends in Fig. 5 are studied at a compressed temperature of $T_C \sim 707$ K and 6% oxygen in the mixture, with varying R_b. The corresponding *n*-butanol/*iso*-octane experiments in Fig. 6 have a compressed temperature of $T_C \sim 752$ K and 15% oxygen in the mixture. For the *n*-butanol/*n*-heptane blends shown in Fig. 5(a), most of the pressure traces exhibit a prominent first-stage pressure rise, while for the *n*-butanol/*iso*-octane blends in Fig. 6 merely a moderate indication of first-stage ignition is seen. It is noted that the symbols in Figs. 5(a) and 6 are meant for labeling purposes only, the pressure data were acquired at 50 samples per millisecond as mentioned earlier.

A feature common to both sets of pressure traces is the decrease in the first-stage ignition delay (τ_1) with decreasing R_b, as demonstrated in Figs. 7(a) and 8(a). However, the total ignition delays (τ) show interesting differences between the two cases. While the total ignition delays shown in Fig. 7(b) for *n*-butanol/*n*-heptane blends increase monotonically with increasing R_b, the total ignition delays for *n*-butanol/*iso*-octane blends in Fig. 8(b) first decrease and then increase slightly as the proportion of *n*-butanol is increased. This experimental trend can be reconciled by noting the fact that the second-stage ignition delay (τ_2) dependence on the proportion of *n*-butanol in the fuel blend (R_b) is completely opposite for the two cases. Note that since the total ignition delay is just the sum of these two ignition delays, $\tau = \tau_1 + \tau_2$, the second-stage ignition delay is the difference between τ and τ_1 in Figs. 7 and 8. Figure 9 demonstrates that the second-stage ignition delays for the blends of *n*-butanol and *n*-heptane increase as the *n*-butanol content is increased, but decreases for the blends of *n*-butanol and *iso*-octane with increasing R_b. The two ignition delays, τ_1 and τ_2, move in the same direction for the *n*-butanol/*n*-heptane blends, but in the opposite direction for the *n*-butanol/*iso*-octane blends. Further investigation is warranted in order to identify the controlling chemistry accounting for the observed trends. In order to predict this type of behavior, the neat components models and the mixture model need to be accurate to within about ±10% in total ignition delay time. Currently, chemical kinetic models are usually considered good if accurate within about ±50%. Higher fidelity models with more accurate rate constants are needed to predict this type of mixture behavior. To aid in this effort, the rate constants that exhibit high sensitivity are identified later in the paper. Once more accurate rate constants are obtained, they need to be implemented in the model and tested using fundamental experimental data, such as the RCM data presented herein.

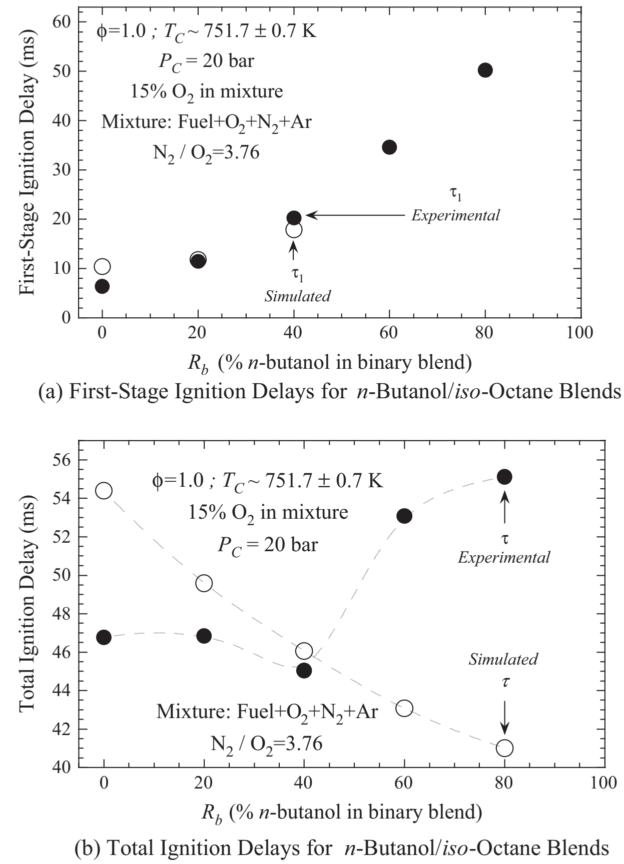
The increasing trend of the first stage ignition delay with the addition of butanol into pure *n*-heptane and *iso*-octane is captured by the simulations as can be seen from the empty symbols in Figs. 7(a) and 8(a). The effect of *n*-butanol addition on the total ignition delay of *n*-butanol/*n*-heptane blends is adequately predicted by the current mechanism, as shown in Fig. 7(b). However, simulations using the current mechanism are unable to describe the trend observed experimentally for the *n*-butanol/*iso*-



(a) First-Stage Ignition Delays for *n*-Butanol/*n*-Heptane Blends

(b) Total Ignition Delays for *n*-Butanol/*n*-Heptane Blends

Fig. 7. (a) First-stage and (b) total ignition delays of *n*-butanol and *n*-heptane blends for varying R_b at a fixed (P_C, T_C, ϕ).



(a) First-Stage Ignition Delays for *n*-Butanol/*iso*-Octane Blends

(b) Total Ignition Delays for *n*-Butanol/*iso*-Octane Blends

Fig. 8. (a) First-stage and (b) total ignition delays of *n*-butanol and *iso*-octane blends for varying R_b at a fixed (P_C, T_C, ϕ).

octane blends, in terms of the total ignition delay variation shown in Fig. 8(b). Efforts are underway to identify the reasons associated with this discrepancy.

The overall prediction of the ignition delay times, while quite satisfactory, does not give a full picture of the aspects that need to be accounted for by a chemical kinetic scheme in the low-to-intermediate temperature region unique to this study. Indeed, the distinguishing factor in this study, when compared to high temperature ignition delays, is the ability to observe features in experimental pressure traces such as the first stage pressure rise and the plateau associated with the diminished reactivity during the second-stage induction time. The ability of the mechanism to accurately simulate the aforementioned qualitative features of ignition is demonstrated in Fig. 10. It can be seen that the simulations are successful in predicting the pressure rise associated with the first stage ignition as well as the reduced reactivity associated with the second stage delay (τ_2), for both *n*-butanol/*n*-heptane and *n*-butanol/*iso*-octane blends. In fact, it is encouraging to find that the simulations very accurately mimic the rather gentle pressure rise combined with the extended second stage ignition delay for *n*-butanol/*iso*-octane blends. That a zero-dimensional RCM simulation is able to reproduce the experimental trend shown in Fig. 10, particularly for a fuel blend involving two different classes of fuels (alkane and alcohol), is also quite encouraging.

Continuing our examination of the trends observed in the current experiments, we seek to explore the combined effect of the variables relevant to the ignition delays for the case where blending has a significant influence. As noted earlier, the relative magnitude of the influence of *n*-butanol blending on ignition delays is much stronger for *n*-heptane when compared to *iso*-octane. Hence, we investigate only the blends of *n*-butanol with *n*-heptane. For the *n*-butanol/*n*-heptane blends we can summarize the influence of blending ratios, at a fixed pressure of P_C = 20 bar and a fixed equivalence ratio of $\phi = 0.4$ in air, by means of an empirical correlation having a form of:

$$\tau = \tau_1 + \tau_2 = A(X_{But})^a (X_{Hep})^b exp(T_a/T_c).$$

The above form is fitted to all data that exhibit a monotonic variation of the ignition delay with temperature, i.e. excluding the NTC region and the high temperature region beyond it. This corresponds to a temperature range of 638–744 K for the current *n*-butanol/*n*-heptane dataset. In this form of the correlation, ignition delay and T_C are in units of *ms* and *K*, respectively, *A* is a constant (with the appropriate units), X_{But} and X_{Hep} are the mole fractions of *n*-butanol and *n*-heptane in the reactive mixture, respectively, *a* is the *n*-butanol exponent, *b* the *n*-heptane exponent, and T_a the activation temperature in units of *K*. The result of such a fitting and the parameters of the resulting fit are shown in Fig. 11(a), while Fig. 11(b) shows the experimental dataset for different blending ratios that were used to form the correlation. As seen from Fig 11(a), the total ignition delay in the empirical correlation is more sensitive to the *n*-heptane mole fraction than the *n*-butanol mole fraction, as seen by the magnitude of the respective exponents. The comparison in Fig. 11(b) shows that the empirical correlation fits well the experimental data over the targeted temperature range.

To better understand the ignition response observed in the simulations, a systematic two-step analysis of the three base mechanisms as well as the resultant *n*-butanol/PRF mechanism is carried out. A brute force sensitivity analysis with respect to the

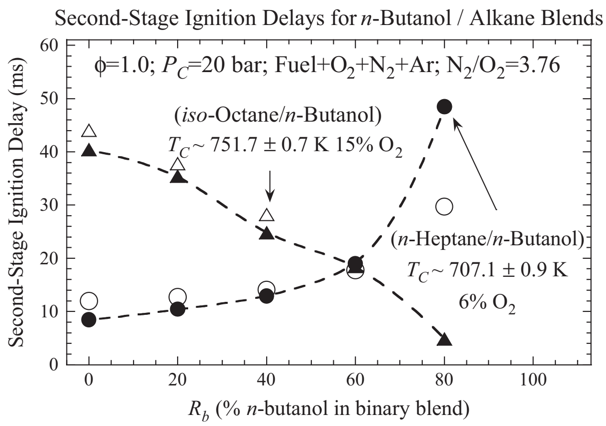


Fig. 9. Dependence of second-stage ignition delays for *n*-butanol/*n*-heptane and *n*-butanol/*iso*-octane blends on the proportion of *n*-butanol in binary fuel blend. Filled and empty symbols have been used for experimental and simulated values, respectively.

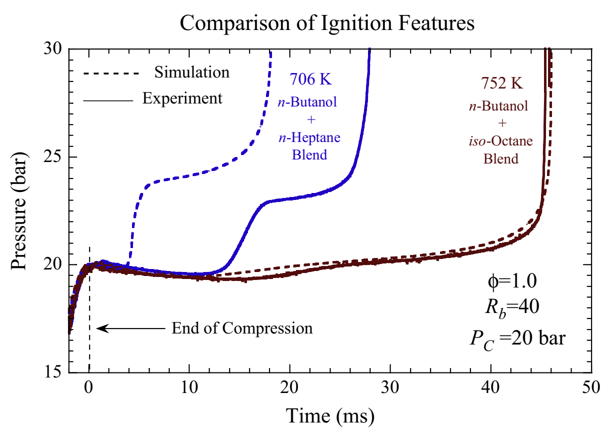


Fig. 10. Demonstration of the capability of the merged mechanism to reproduce the qualitative and quantitative features of the ignition development process.

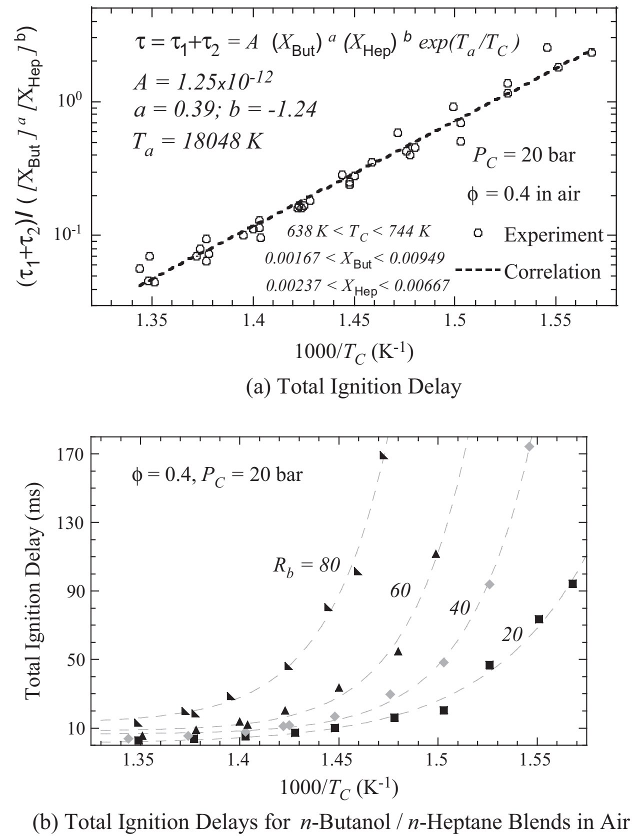


Fig. 11. (a) Total Ignition delay correlation for *n*-butanol/*n*-heptane blends and (b) the experimental data used for correlation in (a). The lower and upper 95% limits for the fit parameters are: $ln(A) \rightarrow (-31.36, -23, 46)$; $a \rightarrow (0.1341, 0.6449)$; $b \rightarrow (-1.6665, -0.8048)$; $T_a \rightarrow (16960, 19135)$.

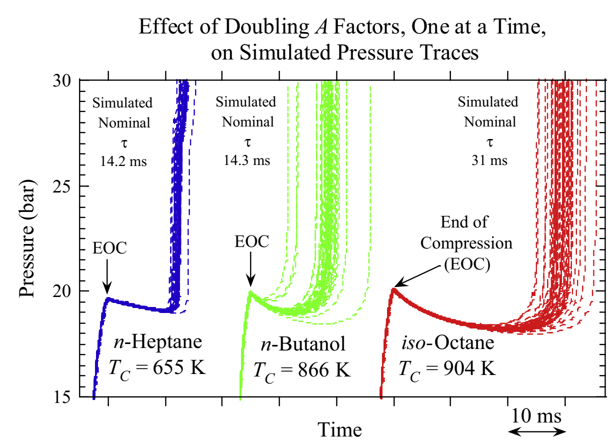
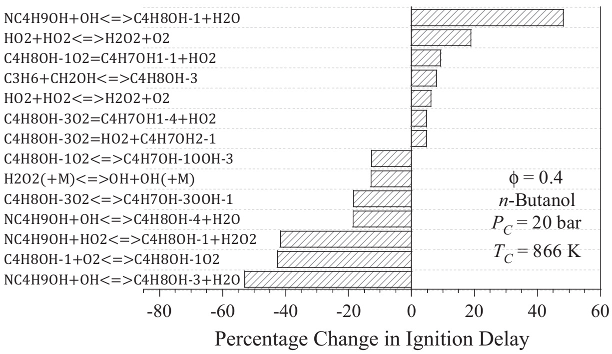


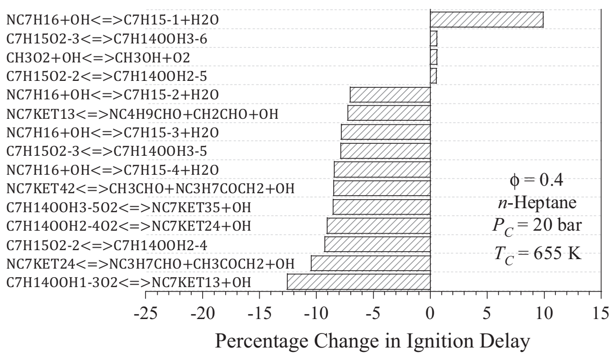
Fig. 12. Effect of brute-force doubling of rate constants on simulated pressure profiles for neat constituent fuels.

pre-exponential factors for the three base mechanisms for *n*-butanol, *n*-heptane, and *iso*-octane is conducted for three separate experimental conditions investigated in this work. The three conditions are chosen to be representative of typical ignition delay times observed in an RCM. The results of such an analysis, which involves doubling of the rate constants, on the RCM pressure evolution profile, are shown in Fig. 12. It can be seen from Fig. 12 that the widest variation in the ignition delay times for a similar one-at-a time variation in pre-exponential factors is observed for *n*-butanol followed by *iso*-octane and *n*-heptane. The scatter shown in Fig. 12 is further quantified and attributed to individuals reactions in the bar plots shown in Fig. 13. A negative percentage change in Fig. 13 implies a higher reactivity or a shorter ignition delay by doubling the corresponding rate constant. It is interesting to note that for each neat fuel component the reactions involving species with the same number of carbon atoms as the parent fuel are found to be important. Note that the temperature conditions corresponding to the analysis for *n*-butanol (T_C = 866 K) and *iso*-octane (T_C = 904 K) are more than 200 K higher than that for *n*-heptane (T_C = 655 K). Therefore, not surprisingly, H_2O_2 decomposition features as one of the contributors in enhancing the reactivity of the system for *n*-butanol and *iso*-octane under these conditions. The H abstraction reactions are also among the most sensitive ones for pure *n*-butanol as well as *iso*-octane.

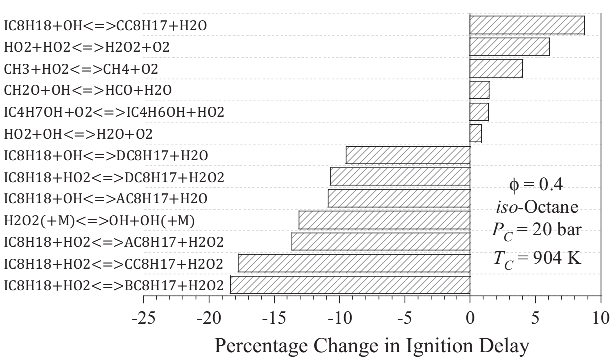
The most sensitive reactions for *n*-butanol are the H abstraction reactions by OH at the α and γ sites. Figure 13(a) shows that increasing the rate constant at the α and β sites by a factor of 2 increases and decreases the ignition delay time by about 50%, respectively. The rate constants used for the α and γ sites taken from Sarathy et al. [65] are shown in Fig. 14 along with the most recent rate constant evaluations of Seal et al. [54] and McGillen et al. [126]. At 866 K, there is about a large factor of 3 uncertainty in the rate constant for the γ site and a relatively small factor of 1.3 for the α site. Further investigations are needed to reduce the uncertainties in these site specific rate constants. Also for *n*-



(a) Influence of Doubling Rate Constants on Simulated Ignition Delays



(b) Influence of Doubling Rate Constants on Simulated Ignition Delays



(c) Influence of Doubling Rate Constants on Simulated Ignition Delays

Fig. 13. Sensitivity of the ignition delay times in the base mechanisms to brute-force doubling of rate constants, based on the results shown in Fig. 12.

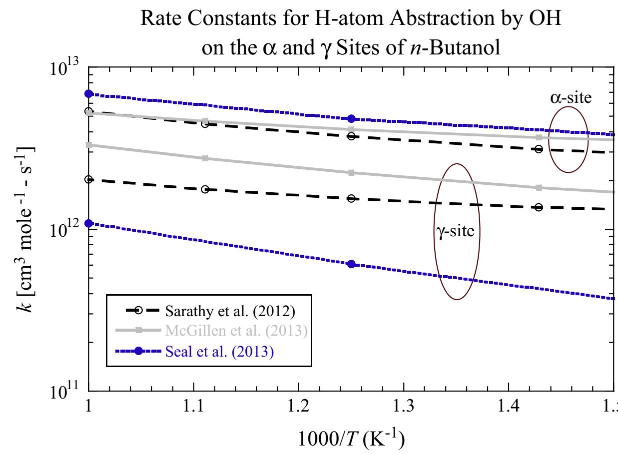
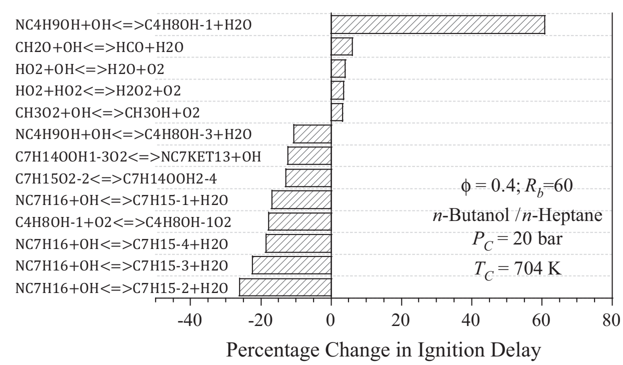
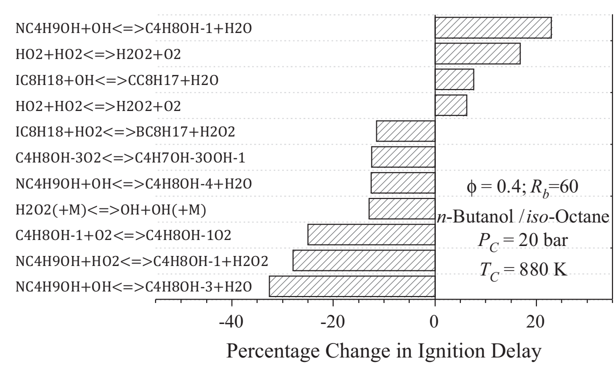


Fig. 14. Comparison of rate constants for H-atom abstraction by OH on the α and γ sites of n-butanol. The rate constant used in the current mechanism is taken from Sarathy et al. [65]. The other rate constants are from McGillen et al. [126] and Seal et al. [54].



(a) Influence of Doubling Rate Constants on Simulated Ignition Delays



(b) Influence of Doubling Rate Constants on Simulated Ignition Delays

Fig. 15. Sensitivity of the ignition delay times in the merged mechanism to brute-force doubling of rate constants for (a) an n-butanol/n-heptane blend at P_C = 20 bar and T_C = 704 K and (b) an n-butanol/iso-octane blend at P_C = 20 bar and T_C = 880 K.

butanol, the $HO_2 + HO_2 = H_2O_2 + O_2$ reaction exhibits a high positive sensitivity. Note that it shows up twice because its complex rate constant behavior requires a specification of two rate constants. This reaction rate constant has recently been measured by Hong et al. [127], which is lower at the present condition than the previously recommended rate constant from Hippler et al. [128] used in the mechanism. Changing this rate constant affects the model predictions of ignition delay for almost all fuel components of interest.

At relatively lower temperatures, the analysis for n-heptane shows the reactions involving RO_2 isomerization and the chain branching ketohydroperoxide decomposition to be the most sensitive. Similar sensitivity results for the blends are shown in Fig. 15 for $R_b = 60$. It can be seen that the H abstraction reactions from the most reactive fuel constituent are among the most sensitive for the blends. The reaction leading to the formation of α-hydroxylbutylperoxy radical is found to be sensitive for both the blends. We note that though the compressed temperature conditions are quite different for Figs. 13(a)–(c), they are under conditions of similar ignition delays. Such conditions are of interest in engine combustion where the ignition delay time is of great significance, and it is important to understand the controlling chemistry for different fuels under similar ignition delay times, for better combustion phasing and control.

One of the reasons to perform the sensitivity analysis is to identify sensitive reactions that are unique to the blends or the neat component models. These reactions could be further investigated to make possible improvements in the model. From the sensitivity results in Figs. 13 and 15, no reactions are identified that are significantly more sensitive for the mixture than for the neat components. It is likely that a combination of reaction rate constants will need to be changed to affect the behavior of the mixture in the desired manner to improve agreement with experiments.

A reaction path analysis for *n*-butanol/*iso*-octane blend of $\phi = 0.4$ in air and $R_b = 60$ at a constant pressure of 20 bar has been conducted using the Cantera package [129]. The results of such an analysis are visualized as graph layouts using NetworkX [130], as shown in Fig. 16. Figure 16(a) and (b) are based on a relative flux threshold value of 0.05 and represent the net flow at the time instant corresponding to a 5 K rise above the initial temperature of 880 K. The thicker end of the connecting line between any two species designates the destination. Subject to the above-mentioned threshold, the *n*-butanol and *iso*-octane fuels are initially two independent pathways with minimal contribution to the common reactive species. It is found that the consumption of *n*-butanol is mainly on account of its reaction with the OH radical leading to hydrogen abstraction from the α, β, γ, and δ carbons (cf. Fig. 1) in the relative proportion of 1, 0.231, 0.385, and 0.340, respectively. In addition, the hydrogen atom abstraction from the 1°, 2°, and 3° sites of *iso*-octane by the OH radical is the major fuel consumption pathway under low temperature conditions. The relative amounts for sites labeled 1, 2, 3, and 4 (cf. Fig. 1) are 0.713, 0.421, 0.490, and 0.482, respectively. Based on the same cutoff, the pathway analysis shows that both *iso*-octane and *n*-butanol begin contributing to the buildup of common radical species at around 892 K, ~12 K rise above the initial temperature. The analysis further identifies C_3H_6 and CH_3 as two key species to which both the parent fuels make a significant common contribution in the intermediate temperature range of 900–1000 K. It is expected that the main chemical interaction between *n*-butanol and *iso*-octane is through small radical species like OH and HO_2, based on this reaction path analysis and the sensitivity analysis in Fig. 15.

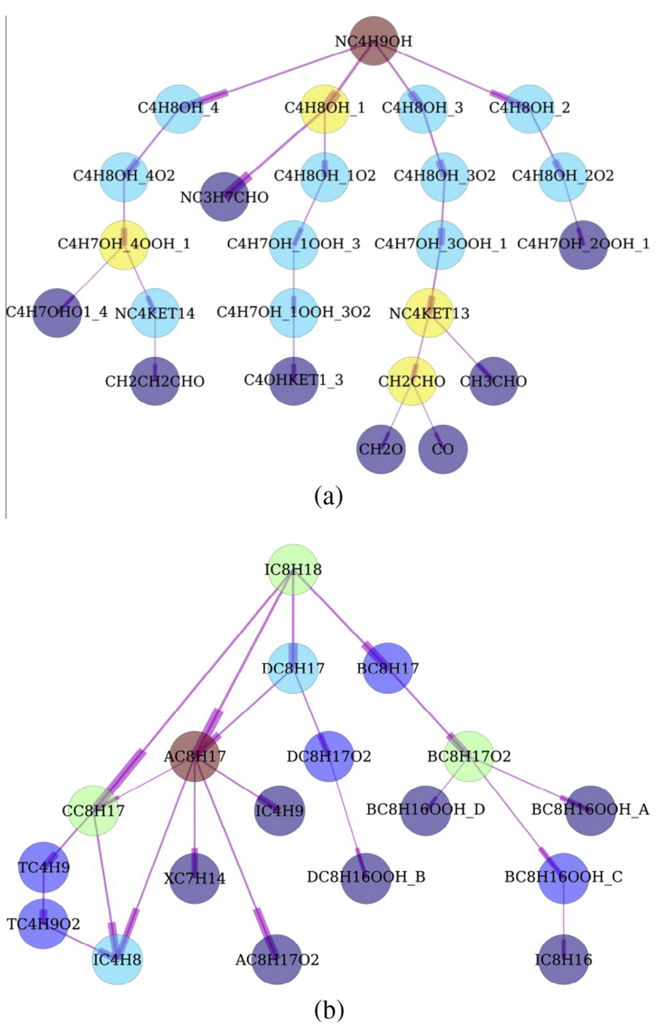


Fig. 16. Flux of the species following the carbon atom for the merged mechanism under constant pressure of 20 bar and initial temperature of 880 K, for an *n*-butanol/*iso*-octane blend of $\phi = 0.4$ and $R_b = 60$. The species naming convention was derived from the three original mechanisms [65,51,110–113] that were used to obtain this merged mechanism.

5. Conclusions

Fundamental ignition data from a rapid compression machine are presented for *n*-butanol blended with key gasoline components, including *n*-heptane and *iso*-octane. This fundamental database provides a reliable validation set for chemical kinetic models of *n*-butanol with *n*-heptane and *iso*-octane. Additionally, the experimental results give insight into how *n*-butanol affects the autoignition chemistry of *n*-heptane and *iso*-octane which are representative of the *n*-alkane and *iso*-alkane chemical classes in gasoline. The present ignition data show that the addition of *n*-butanol to *n*-heptane leads to a decrease in overall reactivity as manifested by the increase in both the first-stage and the total ignition delays for all the conditions studied in this work. The addition of *n*-butanol to *iso*-octane is found to lead to shorter total ignition delays under fuel lean conditions ($\phi = 0.4$) important for HCCI combustion. Ignition response is also studied under stoichiometric conditions by keeping the charge temperature at the end of compression approximately constant while varying R_b, where an unusual behavior is seen when a clear two-stage ignition is present. Under these conditions, addition of *n*-butanol to *iso*-octane decreases the second-stage ignition delay time (τ_2) while addition to *n*-heptane increases τ_2.

An *n*-butanol/PRF mechanism is also generated by a systematic merger of previously reported chemical kinetic schemes for individual constituents in the literature. This mechanism is found to generally well predict the qualitative and quantitative trends for the autoignition of the binary fuel blends studied in this work, while some discrepancies between experiments and simulations for *n*-butanol/*iso*-octane blends are noted. Further experimental and computational studies to explain the underlying chemistry behind the observations are needed, including a possibility of a comprehensive review of the three base chemical kinetic models to account for interaction between fuel constituents. The present new measurements represent targets for the next generation of chemical kinetic models which will require more accurate rate constants and thermodynamic properties.

Acknowledgments

This material is based upon work supported as part of the Combustion Energy Frontier Research Center, an Energy Frontier Research Center funded by the U.S. Department of Energy, Office of Science, Office of Basic Energy Sciences under Award Number DE-SC0001198, and by the National Science Foundation under Grant No. CBET-1402231. The material from LLNL is based upon work supported by the U.S. Department of Energy, Office of Science, Office of Basic Energy Sciences (Wade Sick, program manager) and performed under the auspices of the U.S. Department of Energy by Lawrence Livermore National Laboratory under Contract DE-AC52-07NA27344.

Please cite this article in press as: K. Kumar et al., Combust. Flame (2015), http://dx.doi.org/10.1016/j.combustflame.2015.02.014

Appendix A. Supplementary material

Supplementary data including the chemical kinetic model and the volume-time histories used in the simulations are available for online access with the journal article. Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.combustflame.2015.02.014.

References

[1] B.G. Harvey, H.A. Meylemans, J. Chem. Technol. Biot. 86 (1) (2011) 2–9.
[2] P. Durre, Curr. Opin. Biotechnol. 22 (3) (2011) 331–336.
[3] P.H. Pfromm, V. Amanor-Boadu, R. Nelson, P. Vadlani, R. Madl, Biomass Bioenergy 34 (4) (2010) 515–524.
[4] K.R. Szulczyk, Int. J. Energy Environ. 3 (2010) 501–511.
[5] A. Chotwichien, A. Luengnaruemitchai, S. Jai-In, Fuel 88 (9) (2009) 1618–1624.
[6] P.G. Aleiferis, J. Serras-Pereira, A. Augoye, T.J. Davies, R.F. Cracknell, D. Richardson, Int. J. Heat Mass Transf. 53 (21–22) (2010) 4588–4606.
[7] E. Christensen, J. Yanowitz, M. Ratcliff, R.L. McCormick, Energy Fuels 25 (10) (2011) 4723–4733.
[8] V.F. Andersen, J.E. Anderson, T.J. Wallington, S.A. Mueller, O.J. Nielsen, Energy Fuels 24 (2010) 3647–3654.
[9] V.F. Andersen, J.E. Anderson, T.J. Wallington, S.A. Mueller, O.J. Nielsen, Energy Fuels 24 (2010) 2683–2691.
[10] R. Hanson, S. Kokjohn, D. Splitter, R. Reitz, SAE Int. J. Engines 4 (1) (2011) 394–411.
[11] S. Kokjohn, R. Hanson, D. Splitter, J. Kaddatz, R. Reitz, SAE Int. J. Engines 4 (1) (2011) 360–374.
[12] S.L. Kokjohn, R.M. Hanson, D.A. Splitter, R.D. Reitz, Int. J. Engine Res. 12 (3) (2011) 209–226.
[13] D. Splitter, R. Hanson, S. Kokjohn, R. Reitz, SAE Technical Paper 2011–01–0363, 2011.
[14] R.D. Reitz, Combust. Flame 160 (1) (2013) 1–8.
[15] S.M. Sarathy, P. Oßwald, N. Hansen, K. Kohse-Höinghaus, Prog. Energy Combust. Sci. 44 (2014) 40–102.
[16] P. Dagaut, C. Togbe, Combust. Sci. Technol. 184 (7–8) (2012) 1025–1038.
[17] P. Saisirirat, C. Togbe, S. Chanchaona, F. Foucher, C. Mounaim-Rousselle, P. Dagaut, Proc. Combust. Inst. 33 (Part 2) (2011) 3007–3014.
[18] C. Togbe, G. Dayma, A. Mze-Ahmed, P. Dagaut, Energy Fuels 24 (2010) 3906–3916.
[19] S.M. Sarathy, M.J. Thomson, C. Togbe, P. Dagaut, F. Halter, C. Mounaim-Rousselle, Combust. Flame 156 (4) (2009) 852–864.
[20] P. Dagaut, C. Togbe, Energy Fuels 23 (7) (2009) 3527–3535.
[21] P. Dagaut, C. Togbe, Fuel 87 (15–16) (2008) 3313–3321.
[22] K. Yasunaga, T. Mikajiri, S.M. Sarathy, T. Koike, F. Gillespie, T. Nagy, J.M. Simmie, H.J. Curran, Combust. Flame 159 (6) (2012) 2009–2027.
[23] I. Stranic, S.H. Pyun, D.F. Davidson, R.K. Hanson, Combust. Flame 159 (11) (2012) 3242–3250.
[24] I. Stranic, D.P. Chase, J.T. Harmon, S. Yang, D.F. Davidson, R.K. Hanson, Combust. Flame 159 (2) (2012) 516–527.
[25] C.M. Rosado-Reyes, W. Tsang, J. Phys. Chem. A 116 (40) (2012) 9825–9831.
[26] R.D. Cook, D.F. Davidson, R.K. Hanson, Int. J. Chem. Kinet. 44 (5) (2012) 303–311.
[27] S. Vranckx, K.A. Heufer, C. Lee, H. Olivier, L. Schill, W.A. Kopp, K. Leonhard, C.A. Taatjes, R.X. Fernandes, Combust. Flame 158 (8) (2011) 1444–1455.
[28] K.A. Heufer, R.X. Fernandes, H. Olivier, J. Beeckmann, O. Rohl, N. Peters, Proc. Combust. Inst. 33 (Part 1) (2011) 359–366.
[29] G. Black, H.J. Curran, S. Pichon, J.M. Simmie, V. Zhukov, Combust. Flame 157 (2) (2010) 363–373.
[30] Y. Zhu, D.F. Davidson, R.K. Hanson, Combust. Flame 161 (3) (2014) 634–643.
[31] J. Zhang, S. Niu, Y. Zhang, C. Tang, X. Jiang, E. Hu, Z. Huang, Combust. Flame 160 (1) (2013) 31–39.
[32] H.F. Jin, Y.Z. Wang, K.W. Zhang, H.S. Guo, F. Qi, Proc. Combust. Inst. 34 (2013) 779–786.
[33] G.A. Chung, B. Akih-Kumgeh, G.M.G. Watson, J.M. Bergthorson, Proc. Combust. Inst. 34 (2013) 831–838.
[34] J. Camacho, S. Lieb, H. Wang, Proc. Combust. Inst. 34 (2013) 1853–1860.
[35] G. Broustail, F. Halter, P. Seers, G. Moreac, C. Mounaim-Rousselle, Fuel 106 (2013) 310–317.
[36] A. Frassoldati, R. Grana, T. Faravelli, E. Ranzi, P. Osswald, K. Kohse-Hoinghaus, Combust. Flame 159 (7) (2012) 2295–2311.
[37] M.S. Agathou, D.C. Kyritsis, Appl. Energy 93 (2012) 296–304.
[38] P.S. Veloo, F.N. Egolfopoulos, Proc. Combust. Inst. 33 (Part 1) (2011) 987–993.
[39] P. Osswald, H. Guldenberg, K. Kohse-Hoinghaus, B. Yang, T. Yuan, F. Qi, Combust. Flame 158 (1) (2011) 2–15.
[40] W. Liu, A.P. Kelley, C.K. Law, Proc. Combust. Inst. 33 (2011) 995–1002.
[41] P.S. Veloo, Y.L. Wang, F.N. Egolfopoulos, C.K. Westbrook, Combust. Flame 157 (10) (2010) 1989–2004.
[42] R. Grana, A. Frassoldati, T. Faravelli, U. Niemann, E. Ranzi, R. Seiser, R. Cattolica, K. Seshadri, Combust. Flame 157 (11) (2010) 2137–2154.
[43] B. Yang, P. Osswald, Y.Y. Li, J. Wang, L.X. Wei, Z.Y. Tian, F. Qi, K. Kohse-Hoinghaus, Combust. Flame 148 (4) (2007) 198–209.
[44] H. Jin, A. Cuoci, A. Frassoldati, T. Faravelli, Y. Wang, Y. Li, F. Qi, Combust. Flame 161 (3) (2014) 657–670.
[45] H. Ghiassi, P. Toth, J.S. Lighty, Combust. Flame 161 (3) (2014) 671–679.
[46] F. Wu, C.K. Law, Combust. Flame 160 (12) (2013) 2744–2756.
[47] H. Wang, R. Deneys Reitz, M. Yao, B. Yang, Q. Jiao, L. Qiu, Combust. Flame 160 (3) (2013) 504–519.
[48] J.H. Cai, L.D. Zhang, F. Zhang, Z.D. Wang, Z.J. Cheng, W.H. Yuan, F. Qi, Energy Fuels 26 (9) (2012) 5550–5568.
[49] F.M. Haas, A. Ramcharan, F.L. Dryer, Energy Fuels 25 (9) (2011) 3909–3916.
[50] D.M. Karwat, S.W. Wagnon, P.D. Teini, M.S. Wooldridge, J. Phys. Chem. A 115 (19) (2011) 4909–4921.
[51] D.M.A. Karwat, S.W. Wagnon, M.S. Wooldridge, C.K. Westbrook, J. Phys. Chem. A 116 (51) (2012) 12406–12421.
[52] B.W. Weber, K. Kumar, Y. Zhang, C.J. Sung, Combust. Flame 158 (5) (2011) 809–819.
[53] B.W. Weber, C.J. Sung, Energy Fuels 27 (3) (2013) 1688–1698.
[54] P. Seal, G. Oyedepo, D.G. Truhlar, J. Phys. Chem. A 117 (2) (2013) 275–282.
[55] D. Katsikadakos, C.W. Zhou, J.M. Simmie, H.J. Curran, P.A. Hunt, Y. Hardalupas, A.M.K.P. Taylor, Proc. Combust. Inst. 34 (2013) 483–491.
[56] C.W. Zhou, J.M. Simmie, H.J. Curran, Int. J. Chem. Kinet. 44 (3) (2012) 155–164.
[57] P. Seal, E. Papajak, D.G. Truhlar, J. Phys. Chem. Lett. 3 (2) (2012) 264–271.
[58] D. Katsikadakos, Y. Hardalupas, A.M.K.P. Taylor, P.A. Hunt, Phys. Chem. Chem. Phys. 14 (27) (2012) 9615–9629.
[59] I.M. Alecu, J.J. Zheng, E. Papajak, T. Yu, D.G. Truhlar, J. Phys. Chem. A 116 (50) (2012) 12206–12213.
[60] P. Zhang, S.J. Klippenstein, C.K. Law, J. Phys. Chem. A 117 (9) (2013) 1890–1906.
[61] P. Seal, E. Papajak, T. Yu, D.G. Truhlar, J. Chem. Phys. 136 (3) (2012).
[62] E. Papajak, P. Seal, X.F. Xu, D.G. Truhlar, J. Chem. Phys. 137 (10) (2012).
[63] M.R. Harper, K.M. Van Geem, S.P. Pyl, S.S. Merchant, G.B. Marin, W.H. Green, Combust. Flame 158 (10) (2011). 2075-2075.
[64] J.T. Moss, A.M. Berkowitz, M.A. Oehlschlaeger, J. Biet, V. Warth, P.A. Glaude, F. Battin-Leclerc, J. Phys. Chem. A 112 (43) (2008) 10843–10855.
[65] S.M. Sarathy, S. Vranckx, K. Yasunaga, M. Mehl, P. Osswald, W.K. Metcalfe, C.K. Westbrook, W.J. Pitz, K. Kohse-Hoinghaus, R.X. Fernandes, H.J. Curran, Combust. Flame 159 (6) (2012) 2028–2055.
[66] J. Serras-Pereira, P.G. Aleiferis, D. Richardson, Combust. Sci. Technol. 185 (3) (2013) 484–513.
[67] A. Cairns, H. Zhao, A. Todd, P. Aleiferis, Fuel 106 (2013) 802–813.
[68] C. Tornatore, L. Marchitto, G. Valentino, F.E. Corcione, S.S. Merola, Energy 45 (1) (2012) 277–287.
[69] X.L. Gu, Z.H. Huang, J. Cai, J. Gong, X.S. Wu, C.F. Lee, Fuel 93 (1) (2012) 611–617.
[70] Y.Y. Wu, B.C. Chen, H.C. Tsai, T.C. Liu, Proc. Inst. Mech. Eng. Part a–J. Power Energy 225 (A5) (2011) 579–590.
[71] S. Szwaja, J.D. Naber, Fuel 89 (7) (2010) 1573–1582.
[72] F.N. Alasfour, Int. J. Energy Res. 21 (1) (1997) 21–30.
[73] R.W. Rice, A.K. Sanyal, A.C. Elrod, R.M. Bata, J. Eng. Gas Turb. Power-Trans. Asme 113 (3) (1991) 377–381.
[74] C. Cooney, T. Wallner, S. McConnell, J.C. Gillen, C. Abell, S.A. Miers, J.D. Naber, Effects of Blending Gasoline With Ethanol and Butanol on Engine Efficiency and Emissions Using a Direct-Injection, Spark-Ignition Engine, ASME 2009 Internal Combustion Engine Division Spring Technical Conference, ASME, Milwaukee, 2009, pp. 157–165.
[75] E. Sukjit, J.M. Herreros, K.D. Dearn, A. Tsolakis, K. Theinnoi, Int. J. Hydrogen Energy 38 (3) (2013) 1624–1635.
[76] V. Soloiu, M. Duggan, S. Harp, B. Vlcek, D. Williams, Energy 52 (2013) 143–154.
[77] L. Siwale, L. Kristof, T. Adam, A. Bereczky, M. Mbarawa, A. Penninger, A. Kolesnikov, Fuel 107 (2013) 409–418.
[78] X.L. Gu, G. Li, X. Jiang, Z.H. Huang, C.F. Lee, P I Mech. Eng. D–J. Aut. 227 (2) (2013) 261–271.
[79] G.S. Chen, Y.G. Shen, Q.C. Zhang, M.F. Yao, Z.Q. Zheng, H.F. Liu, Energy 54 (2013) 333–342.
[80] Q.C. Zhang, M.F. Yao, Z.Q. Zheng, H.F. Liu, J. Xu, Energy 47 (1) (2012) 515–521.
[81] G. Valentino, F.E. Corcione, S.E. Iannuzzi, S. Serra, Fuel 92 (1) (2012) 295–307.
[82] G. Valentino, F.E. Corcione, S.E. Iannuzzi, Int. J. Engine Res. 13 (3) (2012) 199–215.
[83] E. Sukjit, J.M. Herreros, K.D. Dearn, R. Garcia-Contreras, A. Tsolakis, Energy 42 (1) (2012) 364–374.
[84] S.L. Lin, W.J. Lee, C.F.F. Lee, Y.P. Wu, Fuel 93 (1) (2012) 364–372.
[85] D.C. Rakopoulos, C.D. Rakopoulos, R.G. Papagiannakis, D.C. Kyritsis, Fuel 90 (5) (2011) 1855–1867.
[86] C.D. Rakopoulos, D.C. Rakopoulos, E.G. Giakoumis, D.C. Kyritsis, Proc. Inst. Mech. Eng. Part a–J. Power Energy 225 (A3) (2011) 289–308.
[87] C.D. Rakopoulos, A.M. Dimaratos, E.G. Giakoumis, D.C. Rakopoulos, Appl. Energy 88 (11) (2011) 3905–3916.
[88] F. Lujaji, L. Kristof, A. Bereczky, M. Mbarawa, Fuel 90 (2) (2011) 505–510.
[89] O. Dogan, Fuel 90 (7) (2011) 2467–2472.
[90] M.F. Yao, H. Wang, Z.Q. Zheng, Y. Yue, Fuel 89 (9) (2010) 2191–2201.
[91] D.C. Rakopoulos, C.D. Rakopoulos, E.G. Giakoumis, A.M. Dimaratos, D.C. Kyritsis, Energy Convers. Manage. 51 (10) (2010) 1989–1997.
[92] M. Karabektas, M. Hosoz, Renew. Energy 34 (6) (2009) 1554–1559.
[93] B.Q. He, M.B. Liu, J. Yuan, H. Zhao, Fuel 108 (2013) 668–674.

[94] S. Visakhamoorthy, J.Z. Wen, S. Sivoththaman, C.R. Koch, Appl. Energy 94 (2012) 166–173.
[95] P. Saisirirat, F. Foucher, S. Chanchaona, C. Mounaim-Rousselle, Energy Fuels 24 (2010) 5404–5409.
[96] Y. Zhang, K. Kumar, C.J. Sung, Autoignition of binary fuel blends of n-butanol and n-heptane in a rapid compression machine, in: 49th AIAA Aerospace Sciences Meeting including the New Horizons Forum and Aerospace Exposition, American Institute of Aeronautics and Astronautics, Orlando, Florida, 2011; AIAA 2011–92.
[97] J.X. Zhang, S.D. Niu, Y.J. Zhang, C.L. Tang, X. Jiang, E.J. Hu, Z.H. Huang, Combust. Flame 160 (1) (2013) 31–39.
[98] X. Zhang, C. Tang, H. Yu, Q. Li, J. Gong, Z. Huang, Energy Fuels 27 (4) (2013) 2327–2335.
[99] G. Mittal, C.J. Sung, Combust Sci. Technol. 179 (3) (2007) 497–530.
[100] K. Kumar, G. Mittal, C.J. Sung, Combust. Flame 156 (6) (2009) 1278–1288.
[101] K. Kumar, C.J. Sung, Int. J. Chem. Kinet. 43 (4) (2011) 175–184.
[102] T. Tsujimura, W.J. Pitz, F. Gillespie, H.J. Curran, B.W. Weber, Y. Zhang, C.-J. Sung, Energy Fuels 26 (8) (2012) 4871–4886.
[103] S.M. Sarathy, S. Park, B.W. Weber, W. Wang, P.S. Veloo, A.C. Davis, C. Togbe, C.K. Westbrook, O. Park, G. Dayma, Z. Luo, M.A. Oehlschlaeger, F.N. Egolfopoulos, T. Lu, W.J. Pitz, C.-J. Sung, P. Dagaut, Combust. Flame 160 (12) (2013) 2712–2728.
[104] G. Mittal, C.J. Sung, Combust. Flame 150 (4) (2007) 355–368.
[105] G. Mittal, C.J. Sung, Combust. Flame 156 (9) (2009) 1852–1855.
[106] G. Mittal, C.J. Sung, Combust. Flame 155 (3) (2008) 431–439.
[107] G. Kukkadapu, K. Kumar, C.J. Sung, M. Mehl, W.J. Pitz, Proc. Combust. Inst. 34 (1) (2013) 345–352.
[108] S. Dooley, S.H. Won, M. Chaos, J. Heyne, Y.G. Ju, F.L. Dryer, K. Kumar, C.J. Sung, H.W. Wang, M.A. Oehlschlaeger, R.J. Santoro, T.A. Litzinger, Combust. Flame 157 (12) (2010) 2333–2339.
[109] S. Dooley, S.H. Won, J. Heyne, T.I. Farouk, Y. Ju, F.L. Dryer, K. Kumar, X. Hui, C.-J. Sung, H. Wang, M.A. Oehlschlaeger, V. Iyer, S. Iyer, T.A. Litzinger, R.J. Santoro, T. Malewicki, K. Brezinsky, Combust. Flame 159 (4) (2012) 1444–1466.
[110] H.J. Curran, P. Gaffuri, W.J. Pitz, C.K. Westbrook, Combust. Flame 129 (3) (2002) 253–280.
[111] M. Mehl, H.J. Curran, W.J. Pitz, C.K. Westbrook, Chemical kinetic modeling of component mixtures relevant to gasoline, in: 4th European Combustion Meeting, Vienna, Austria, 2009; LLNL-CONF-410968.
[112] M. Mehl, W.J. Pitz, M. Sjöberg, J.E. Dec, Detailed kinetic modeling of low-temperature heat release for PRF fuels in an HCCI engine, in: SAE 2009 International Powertrains, Fuels and Lubricants Meeting, Florence, Italy, SAE Paper No. 2009–01-1806.
[113] https://www-pls.llnl.gov/?url=science_and_technology-chemistry-combustion-iso_octane_version_3 (access 22.05.14).
[114] in: CHEMKIN-PRO, Reaction Design: San Diego, 2011.
[115] in: R.J. Kee, F.M. Rupley, E. Meeks, J.A. Miller, CHEMKIN-III: A FORTRAN Chemical Kinetics Package for the Analysis of Gas-Phase Chemical and Plasma Kinetics, 1996.
[116] in: A.E. Lutz, R.J. Kee, J.A. Miller, SENKIN: A FORTRAN Program for Predicting Homogeneous Gas Phase Chemical Kinetics with Sensitivity Analysis, 1988.
[117] W.D. McCain Jr., Properties of Petroleum Fluids, PennWell, 1990.
[118] C.K. Westbrook, Proc. Combust. Inst. 28 (2000) 1563–1577.
[119] M. Carlier, C. Corre, R. Minetti, J.F. Pauwels, M. Ribaucour, L.R. Sochet, Symp. (Int.) Combust. 23 (1) (1991) 1753–1758.
[120] R. Minetti, M. Ribaucour, M. Carlier, C. Fittschen, L.R. Sochet, Combust. Flame 96 (3) (1994) 201–211.
[121] G. da Silva, J.W. Bozzelli, L. Liang, J.T. Farrell, J. Phys. Chem. A 113 (31) (2009) 8923–8933.
[122] R.C. Santana, P.T. Do, M. Santikunaporn, W.E. Alvarez, J.D. Taylor, E.L. Sughrue, D.E. Resasco, Fuel 85 (5–6) (2006) 643–656.
[123] G. Knothe, Prog. Energy Combust. Sci. 36 (3) (2010) 364–373.
[124] R. Daniel, G. Tian, H. Xu, S. Shuai, Fuel 99 (2012) 72–82.
[125] M. Mehl, J.Y. Chen, W.J. Pitz, S.M. Sarathy, C.K. Westbrook, Energy Fuels 25 (11) (2011) 5215–5223.
[126] M.R. McGillen, M. Baasandorj, J.B. Burkholder, J. Phys. Chem. A 117 (22) (2013) 4636–4656.
[127] Z. Hong, K.-Y. Lam, R. Sur, S. Wang, D.F. Davidson, R.K. Hanson, Proc. Combust. Inst. 34 (1) (2013) 565–571.
[128] H. Hippler, J. Troe, J. Willner, J. Chem. Phys. 93 (3) (1990) 1755–1760.
[129] D. Goodwin, An open-source, extensible software suite for CVD process simulation, in: M. Allendorf; F. Maury; F. Teyssandier, (Eds.), Chemical Vapor Deposition XVI and EUROCVD 14, The Electrochemical Society, 2003, pp. 155–162.
[130] A.A. Hagberg, D.A. Schult, P.J. Swart, Exploring network structure, dynamics, and function using NetworkX, in: Proceedings of the 7th Python in Science Conference (SciPy2008), G. Varoquaux; T. Vaught; J. Millman, (Eds.), Pasadena, CA USA, August 2008; 11–15.